TRAINITY PROJECT 8

ABC Call Volume Trend

Project Description: This project focuses on Customer Experience analytics, mainly focusing on data from the inbound calling team. We have to analyse data from the call centre agents. Analysing which Time intervals have the highest volume of calls and coming up with a plan to organize the man-force in a way of maximum efficiency.

Tech-Stack Used: Microsoft Excel

Data Analytics Tasks:

You have been provided with a dataset that contains information about the inbound calls received by a company named ABC, which operates in the insurance sector. Your task is to use this data to answer the following questions:

1. **Average Call Duration:** Determine the average duration of all incoming calls received by agents. This should be calculated for each time bucket.

   Your Task: What is the average duration of calls for each time bucket?

2. **Call Volume Analysis:** Visualize the total number of calls received. This should be represented as a graph or chart showing the number of calls against time. Time should be represented in buckets (e.g., 1-2, 2-3, etc.).

   Your Task: Can you create a chart or graph that shows the number of calls received in each time bucket?

3. **Manpower Planning:** The current rate of abandoned calls is approximately 30%. Propose a plan for manpower allocation during each time bucket (from 9 am to 9 pm) to reduce the abandon rate to 10%. In other words, you need to calculate the minimum number of agents required in each time bucket to ensure that at least 90 out of 100 calls are answered.

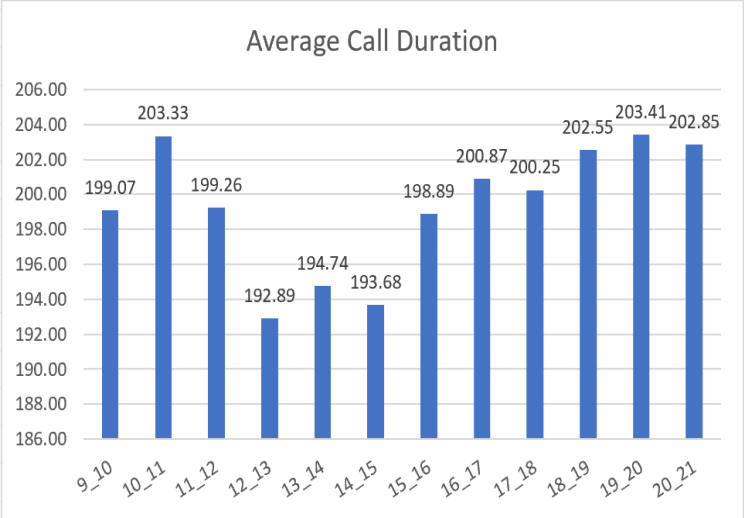
   Your Task: What is the minimum number of agents required in each time bucket to reduce the abandon rate to 10%?

4. **Night Shift Manpower Planning:** Customers also call ABC Insurance Company at night but don't get an answer because there are no agents available. This creates a poor customer experience. Assume that for every 100 calls that customers make between 9 am and 9 pm, they also make 30 calls at night between 9 pm and 9 am. The distribution of these 30 calls is as follows:

   Your Task: Propose a manpower plan for each time bucket throughout the day, keeping the maximum abandon rate at 10%.

Task 1: What is the average duration of calls for each time bucket?

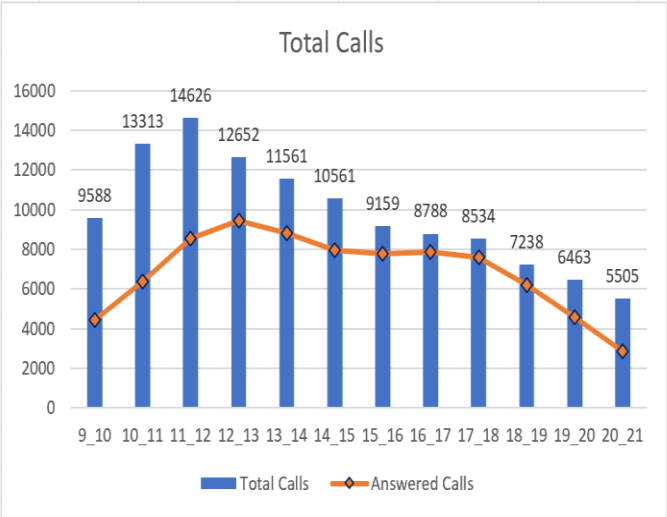
Time Bucket	Avg Duration
9_10	199.07
10_11	203.33
11_12	199.26
12_13	192.89
13_14	194.74
14_15	193.68
15_16	198.89
16_17	200.87
17_18	200.25
18_19	202.55
19_20	203.41
20_21	202.85



- The average duration of a call is the highest in '19_20' time bucket followed closely by '20_21' and '18_19'.
- The lowest average duration of a call is during '12_13' time bucket followed by '14_15' and '13_14'. These are the lunch hours for offices which would conclude the comparatively lower call durations as the calls might have been abandoned by the agents.

Task 2: Visualize the total number of calls received. This should be represented as a graph or chart showing the number of calls against time. Time should be represented in buckets (e.g., 1-2, 2-3, etc.).

Time Bucket	Total Calls	Answered Calls
9_10	9588	4428
10_11	13313	6368
11_12	14626	8560
12_13	12652	9432
13_14	11561	8829
14_15	10561	7974
15_16	9159	7760
16_17	8788	7852
17_18	8534	7601
18_19	7238	6200
19_20	6463	4578
20_21	5505	2870



- The highest number of calls were received in the '11_12' time bucket followed by 10_11 and 12_13. Whereas the lowest number of calls were received in the '20_21' time bucket followed by '19_20' and '18_19'.
- From the previous task we noticed that during the high call volume hours the average time per call was comparatively lower. That finding is justified by this graph as the orange line quantifies the number of answered call out of the calls received.
- The orange line can be seen at much lower number for high volume call time buckets. This explains a non-functional man-force allocation within the team.

Task 3: The current rate of abandoned calls is approximately 30%. Propose a plan for manpower allocation during each time bucket (from 9 am to 9 pm) to reduce the abandon rate to 10%. In other words, you need to calculate the minimum number of agents required in each time bucket to ensure that at least 90 out of 100 calls are answered.

Total Agents are 65				
Total minutes on 01-01-22	01-01-2022	676664	187.9622	42
	02-01-2022	574003	159.4453	
	03-01-2022	812863	225.7953	

Assuming an agent works 9 hours a day with 1.5 hours in breaks leaving 7.5 working hours.
If an agent is on call 60% of the time, that puts them on call for 4.5 hours a day.

Agents	Efficiency
42	70
x	90
x =	54

Time Bucket	Call Seconds	Call Hours	Contribution	Agents Required
9_10	35313	9.81	0.05	3
10_11	53087	14.75	0.08	4
11_12	67751	18.82	0.10	5
12_13	72680	20.19	0.11	6
13_14	59693	16.58	0.09	5
14_15	76137	21.15	0.11	6
15_16	65689	18.25	0.10	5
16_17	59464	16.52	0.09	5
17_18	68155	18.93	0.10	5
18_19	53096	14.75	0.08	4
19_20	40141	11.15	0.06	3
20_21	25458	7.07	0.04	2
Total	676664	187.96	1.00	54

- First, the total number of calling hours for 01-01-2022 was calculated. This number was then divided by 4.5 (time spent by an agent on call).
- Therefore, we got 42 agents working when the efficiency was 70%, now for an efficiency of 90%, the agents required were found to be 54.
- Then, the call hours were calculated for each time bucket on 01-01-2022 and their contributions were calculated to total call hours. These contributions were then multiplied by 54 (total agents required) to get the final allocation.
- The most agents were required in '12_13' and '14_15' time buckets due to high call volumes and high call abandon rates.

Task 4: Customers also call ABC Insurance Company at night but don't get an answer because there are no agents available. This creates a poor customer experience. Assume that for every 100 calls that customers make between 9 am and 9 pm, they also make 30 calls at night between 9 pm and 9 am. The distribution of these 30 calls is as follows:

Distribution of 30 calls coming in night for every 100 calls coming in between 9am - 9pm (i.e. 12 hrs slot)											
9pm- 10pm	10pm - 11pm	11pm- 12am	12am- 1am	1am - 2am	2am - 3am	3am - 4am	4am - 5am	5am - 6am	6am - 7am	7am - 8am	8am - 9am
3	3	2	2	1	1	1	1	3	4	4	5

Date	Calls	Time	Avg Per Call
01-01-2022	4644	676664	145.70715
02-01-2022	3351	574003	171.29305
03-01-2022	4789	812863	169.73544
04-01-2022	5113	861946	168.57931
05-01-2022	4790	846798	176.78455
06-01-2022	4951	829040	167.449
07-01-2022	4948	757019	152.99495
08-01-2022	4672	735444	157.41524
09-01-2022	3652	541147	148.17826
10-01-2022	4983	778739	156.27915
11-01-2022	4637	785717	169.44512
12-01-2022	4643	709934	152.90416
13-01-2022	4123	691320	167.67402
14-01-2022	3155	564227	178.83582
15-01-2022	3058	556267	181.90549
16-01-2022	5142	674394	131.15403
17-01-2022	22347	945615	42.315076
18-01-2022	5774	796768	137.99238
19-01-2022	4703	750270	159.53009
20-01-2022	4322	759613	175.75497
21-01-2022	3675	639855	174.1102
22-01-2022	3291	621577	188.87177
23-01-2022	3225	553899	171.75163
			158.55047

Average Calls Per Day	5130		
Calls in Night Shift	1539		
		In Seconds	In hours
Average Call Time at Night Shift		244005	68
Work Force required for Night		15	

Time Bucket	Call Distribution	Distribution	Agents
9_10	3	0.10	2
10_11	3	0.10	2
11_12	2	0.07	1
12_13	2	0.07	1
13_14	1	0.03	1
14_15	1	0.03	1
15_16	1	0.03	1
16_17	1	0.03	1
17_18	3	0.10	2
18_19	4	0.13	2
19_20	4	0.13	2
20_21	5	0.17	3
Total	30	1.00	15.00

- The average calls per day was calculated as 5130 and 30% of that is 1539 (average calls per day during night hours).
- Now, the average call duration at night shift was calculated as 68 hours, giving the total work-force required as 15 (dividing hours by 4.5).

- Then 15 was multiplied by the distribution per time bucket to find out the most efficient allocation.
- The highest number of agents were required in the '20_21' time bucket as they would be the waking hours of the customers in turn increasing the call volumes.

Result: Through this project, I learnt about pivot tables and data validation. Along with these, I also learnt about statistics and how to make calculations relevant to the data and the problem statements as the 3rd and 4th tasks required a little bit of problem solving and critical thinkings. I also learnt about how to derive meaningful insights through a graph impacting the business directly.

Excel Link:

https://docs.google.com/spreadsheets/d/1JsIlKHF9ZYwXHDWYZmJAtbyEkRg_uIju/edit?usp=sharing&ouid=100878477496382804997&rtpof=true&sd=true